

After completing this lengthy manuscript of over 400,000 words, I felt a sense of emptiness within me for a long time. I must say that when I first began contemplating the small things around me, I certainly knew that something extraordinary would happen. But at that time, I did not yet realize that social science is a tightly woven web, nor did I know what would happen when an energetic individual, armed with the scientific method, found a loose thread and possessed the determination and perseverance to unravel it layer by layer.

By the time I graduated with my bachelor's degree and left Beijing, I had already accomplished the most extraordinary innovation in my theory: tracing the roots of historical materialism back to the material world itself—namely, identifying which factor of production occupies the core position in labor and production. I knew I had single-handedly broken through the limitation of Marx and Engels, who had merely traced their political economy and class-based view of history back to humanity itself. Thus, I avoided the dilemma of having to resort to moral critique, completely sweeping away this "idealistic materialism." Likewise, it was the proposition of the *primary factor of production* (首位生产要素) that enabled me to provide relatively precise definitions for those ambiguous and erroneous concepts previously utilized by others. Although definitions in social science cannot be as flawlessly precise as in mathematics, I believe we can at least render them capable of serving as the cornerstone for logical reasoning and theoretical paradigms.

Therefore, when I resumed my previous research in Xi'an, I believed this theory was applicable not only to China but must possess a certain universality. Consequently, as I began testing it against other unobserved historical space-times across the world, I anticipated that my theory would achieve good fitting results—at least capable of resolving seventy to eighty percent of the difficult problems, which would already far surpass my predecessors. However, the final result was far more optimistic than I had initially imagined: the affairs of the world in all directions, from ancient times to the present, can be measured by a single yardstick. Finally, as the year 2025 arrived, I felt with increasing intensity in practice the impact that AI is exerting upon human civilization; humanity is even standing at the critical juncture of a historical turning point. In the afterword, I pointed out that although I do not know where the changes I have observed will ultimately lead us, for myself and for the hundreds of millions of my compatriots living under the restoration-bureaucratic system, it is ultimately something to be gratified about.

What about the future, then? Since we know how humanity has arrived at today, we naturally wish to know where humanity will head in the days to come. Even from the perspective of scientific research, this is a perfectly justified question. People must always first study and understand the laws, and then master and utilize these laws to change their own lives. The history of mathematics and the natural sciences verifies this point; if social science can truly be studied like a hard science, it will presumably undergo this same process. Perhaps it will not be until many years later that people will be able to make relatively accurate predictions about the future through known social laws. Just as the classical mechanics system and the method of calculus, for over a century after being pioneered by Newton and others, remained merely intellectual games for the minority and did not profoundly alter the human way of life.

But there is reason to believe that humanity's progression from understanding scientific knowledge to utilizing it is only a matter of time. People might say that even so, it is far too early to be thinking about the future now; we do not even know how long it will take for a paradigm revolution (范式革

命) in social science to be accepted by the public, let alone the fact that its scientification is still in its nascent stage. I myself am also clear that writing these words down now is premature, still too akin to a kind of sci-fi rhapsody. However, as an individual, time waits for no one. I have just turned 25 years old, a time when my energy is most abundant and my thinking most active. If I do not write these thoughts down now, I cannot guarantee that I will have this opportunity and energy in the future. How brief a single person's life is within the history of the entire species!

According to the analytical method we established in the manuscript, the future of human society fundamentally depends on which factor of production can attain the status of the *primary factor of production* (首位生产要素) following capital. Standing at the midpoint of 2026, it seems two factors of production are most likely to be crowned the new king.

The first is **Minzhi** (民智). This concept is not easy to define, and people easily equate it with knowledge, but knowledge is only a part of *Minzhi*. I believe a better definition of *Minzhi* is: knowledge at the rational level, as well as the ability to acquire, understand, and master knowledge; coupled with emotions at the perceptual level, as well as the ability to generate, perceive, and express emotions. The reason *Minzhi* becomes a candidate is that with the increase in human-utilizable computing power, the emergence of advanced AI, and the popularization of production automation, primary labor will lose its value, and the production of material wealth might enter a state where "marginal costs approach zero." This means the capital used to organize production will no longer be scarce. By this time, the truly important questions become "what to do" and "why do it," and *Minzhi* could become the *primary factor of production*—the one making the greatest contribution, situated at the core of organization, and convertible into other factors of production.

Such a future is highly worth looking forward to; it signifies the complete invalidation of the exploitative methods of the **restoration-bureaucratic system** (复辟官僚制度). Rulers can control human bodies, monopolize the output of land, and extort liquid capital, but they can never obtain genuinely creative intellect through extortion. *Minzhi* possesses a strong decentralizing attribute; to maximize its output, society must provide a free, tolerant, and democratic environment. The true greatness of *Minzhi* lies in this: it is immune to exploitation. When violated, it would rather self-destruct than fall into the hands of perpetrators. If an era truly emerges in the future where *Minzhi* serves as the *primary factor of production*, it will be the perfect opportunity to eliminate the suffering of all nations and promote a closer solidarity of humanity.

The other alternative is not nearly as welcome, and that is "Energy." When we mention that energy might become the *primary factor of production*, people might easily think of the famous Kardashev scale, and indeed, there is a connection. For *Minzhi* to be fully unleashed, both the automation of production and the massive computing power required to support it demand staggering energy supplies. This means humanity cannot forever rely solely on all the fossil fuels and nuclear energy on Earth, or even surface solar energy—it will not be enough. We will ultimately need to step out of the cradle of Earth and achieve the acquisition and utilization of energy on an interstellar scale. In a certain sense, energy will replace the status once held by land. In the past, the reason land—as a factor of production (and a means of production) with strong spatial attributes—was contested over by people was largely because it was almost the sole carrier of energy available to ancient humans in the pre-industrial era: land yielded grain, and grain converted into population. When human technological capabilities reach a certain level, perhaps even capable of fully utilizing the energy of

the sun, the two-dimensional spatial limits of land will be broken. With abundant energy, humanity can transform lands that previously could not provide economic output, and even establish settlements in space that are currently unimaginable. Energy, in a sense, will become the three-dimensional "land" of the future, and controlling it means controlling the "power of genesis" (造物权) of the new world.

Unlike *Minzhi*, energy can be monopolized by the state apparatus because it possesses an extremely high threshold for technology and capital, as well as physical concentration. The *Minzhi* of the general public will lose all room for bargaining in the face of energy monopolies, because even computing power, and even the life support systems of their environments, will be indirectly controlled by energy owners. This future, under rather terrible circumstances, would form the "cyberpunk-esque" world we speak of today. But what leaves one finding it difficult to evaluate is that even a monopolized cyberpunk world—under energy as the *primary factor of production*—is far better than a **restoration-bureaucratic state** (复辟官僚国家) like the Ming and Qing empires. Because the operational logic of the latter dictates that it *must* actively torture you, squeeze you, and must use exhaustion, ignorance, and fear to turn every human individual into a dry battery, driving *Minli* (民力) to sustain its own existence. In a cyberpunk era, however, AI and robotics render the extraction of the lower class's *Minli* meaningless; the attitude of the rulers towards the ruled will shift from "cruel extortion" to "callous indifference." Although this still sounds sorrowful, it does physically end that historical cycle (历史周期律) that operated by feeding on the flesh and blood of the living.

How, then, can we pursue a better future and avoid the worse one? Do not forget that we already know scarcity (稀缺性关系) plays a crucial regulatory role in the *relations of production*. The answer is ready to be revealed: to ensure that *Minzhi* fully manifests its power in the next era, we must ensure that energy is not in a state of extreme shortage. Only when energy technology achieves an explosive breakthrough and energy is no longer scarce at all will humanity's focus of competition shift completely to *Minzhi*. Oppression and monopoly will then lose their physical significance, and our species will have the opportunity to usher in a free and comprehensive emancipation. If breakthroughs in energy technology are delayed while AI develops too rapidly, energy will remain in a state of scarcity, thereby carrying the risk of being monopolized.

However, what I want to say is that even if future risks can already be foreseen, it does not mean humanity should adopt a policy of "long-termism" (远视主义). After all, risk is only a possibility, and the future situation cannot be accurately known today. Strengthening control over the entire society for the sake of an "uncertain grand goal" is highly likely to bring about disastrous consequences in the present, because this is exactly what the **restoration-bureaucratic system** (复辟官僚体制) loves to do—just as the Chinese-style values reflected in *The Wandering Earth* demonstrate. Enhancing science popularization for the masses, coupled with policy support and continuous investment and sponsorship in scientific research, can also accelerate the arrival of the new energy revolution. In short, people should adopt moderate methods rather than heavy-handed administrative measures.

Of course, we have no reason to believe that *Minzhi* and energy represent two mutually exclusive paths. In fact, it is highly likely that they will each govern two chronologically sequential eras. First, *Minzhi* becomes the *primary factor of production*. In this era, humanity first crosses the threshold of a Type I civilization on the Kardashev scale. In the continuing evolutionary process toward a Type II civilization, energy will become the *primary factor of production*. This is also a gratifying result, because the era of *Minzhi* will bring humanity yet another round of emancipation, allowing us to proceed into the energy-dominated era of interstellar navigation without having to carry the historical baggage accumulated thus far. People will view politics more rationally and wisely than they do today.

This would even lead to an interesting scenario: if we assume that we are indeed going to experience the two eras of *Minzhi* and energy next, then from ancient times to the future, there will have been eras dominated by five respective *primary factors of production*: primary labor, land, capital, *Minzhi* (in a sense, advanced labor), and energy. If this prediction comes true, then we are indeed at the beginning of the "second stage"—an unquestionable turning point. It is not hard to see that both primary labor and *Minzhi* are creative forces related to human beings themselves, while the subsequent land and energy are both physical attributes of the natural world; and between these two stages, capital served as the "lubricant."

We can even boldly predict that there will very likely be a cosmic-level era of "**The Great**

Expansion" (大扩张) in the future. When humanity pushes into deep space using sub-light speed spacecraft, it will inevitably create a situation similar to the dispersal of late *Homo sapiens*: the overall expansion of the species will be rapid, but communication between individual entities (settlements) will be delayed. Although humanity can rapidly spread its existence across a vast cosmic space through intergenerational relay expansion, continuously utilizing earlier established outposts (for example, taking 10,000 to 1,000,000 years, which is still relatively fast on a geological time scale), it will be difficult for various cosmic settlements to conduct two-way communication. They are highly likely to develop relatively independently. Because of the hard limit of the speed of light, will there be early states (早期国家) on a cosmic scale in the future? Out of the necessity to reduce management costs, will there be "energy feudalization" (能源封建化) on a cosmic scale? If some super-technologies currently beyond our anticipation (such as wormholes) appear in the even more distant future, then the humanity of cosmic-scale energy feudalization will evolve even further. What a magnificent imagination this is!

Finally, there is one more point we can predict with relative certainty: class in the traditional sense is disappearing. This does not mean that classes themselves are vanishing, but rather that maintaining this definition is losing its significance. In describing the past history of humanity, the reason we emphasized the class attributes of people and societies was that different classes occupied different relative positions and undertook different functions in social production, thus generating specific modes of interaction. Specifically, past class societies always had a primary exploiting class and an exploited class. This was because even if a certain group controlled the *primary factor of production* and thus dominated social production, they ultimately still had to utilize *Minli* to carry out labor production. Economic inequality eventually brought about social inequality.

However, with the comprehensive rollout of artificial intelligence and automation, the majority of labor production will no longer be completed by humans; meanwhile, the creative thinking and

emotional activities that humans will continue to be responsible for cannot be obtained through exploitation. This will undoubtedly declare the end of societies characterized by relatively distinct class antagonisms of the past. In the future, people will likely no longer be able to accurately point out an economically ruling class and a ruled class. With the weakening role that class plays in human society, the class transition (阶级更替) that previously supported the rise of empires during transitional periods of social formations will no longer occur.

What impact all these various changes will have on the future development of humanity remains unknown today. But for me, they at least indicate one point: the fact that my manuscript has been written in today's era indeed signifies the end of a long epoch. Perhaps, this can be counted as the will of Heaven.

以下为中文原文：

在完成超过四十万字的漫长文稿之后，我心里在很长时间里都感到一种空虚。我必须要说，在我从身边的小事开始思考的时候，我当然知道会发生不寻常的事情。但是我那时还不知道社会科学是一张编织紧密的网，不知道当精力充沛而又掌握科学方法的人找到一个线头，有决心和毅力抽丝剥茧一样扯开它的时候会发生什么。在我本科毕业离开北京的时候，我已经完成了理论中最为非凡的创新，那就是我将历史唯物主义的根源上溯到物质世界本身，也即哪一种生产要素处于劳动生产中的核心位置。我知道自己一举突破了马克思和恩格斯当年仅仅把其政治经济学和阶级史观上溯到人本身的局限，避免了必须诉诸道德批判的困境，将这种“唯心的唯物主义”一扫而空。同样是首位生产要素的提出，使我能够对于之前人们所运用的那些模糊的，错误的概念进行相对精确的定义。尽管社会科学中的定义无法像数学那样精准无误，但是我想我们至少能够做到让其能够作为一些逻辑推理和理论范式的基石。所以当我在西安重新继续之前的研究时，我相信这个理论不只适用于中国，他一定具有某种普世性。于是我一边开始对于世界上其他没有观察过的历史时空进行检验，一边期待我的理论能够取得良好的拟合结果，至少能够解决七、八成的疑难问题，这就已经远远超过前人了。然而最终的结果比我一开始想的还乐观得多，天下四方，古往今来之事，可以用一根准绳来衡量。最后，当 2025 年到来，我在实践中越来越强烈感受到了 AI 对于人类文明造成的冲击，甚至人类正处于历史转折的紧要关头。在后记中我指出，我观察到的变化虽然不知最终将把我们带向何方，但是对于我自己以及生活在复辟官僚体制下的亿万同胞来说，总归是一件值得欣慰的事。

那么，未来呢？我们既然知道了人类是如何走到今天，自然就想知道人类日后将会走向何处。即使从科学研究的角度看，这也是个理所应当的问题。人们总要先去学习和理解规律，然后再掌握并利用规律去改变自己的生活。数学和自然科学的历史印证了这一点，如果社会科学真的能像科学一样被研究的话，那大概也会经历这个过程。或许直到很多年之后，人们才能通过已知的社会规律对未来进行相对准确的预测。就如同经典力学体系、微积分方法在

被牛顿等人初创的一个多世纪里，仍然只是少数人的智力游戏，并没有深刻改变人类的生活方式。但是有理由相信人类把科学知识从理解到利用是或早或晚的事情。人们或许会说，饶是如此，现在就想未来的事情也太早了，仅仅是社会科学上的范式革命都不知道还要过多长时间才能为人所接受，更不要提其科学化还处于草创阶段。我自己也清楚现在写下这些文字为时尚早，还太接近一种科幻的狂想。然而作为个体，岁月不等待任何人，我现在刚刚 25 周岁，正是精力最充沛，思维最活跃的时候。如果现在不把这些想法写下来，我无法保证日后还有这个机会和精力，一个人的生命在整个物种的历史中是多么短暂啊！

按照我们在文稿中所建立的分析方法，人类社会的未来在根本上取决于在资本之后，哪一种生产要素能够取得首位生产要素的地位。站在 2026 年中这个时间点，看起来有两个生产要素最有可能加冕为新王。第一个是“民智”，这个概念不太好定义，人们很容易将它与知识画等号，但知识只是“民智”的一部分。我认为对民智的比较好的定义是：理性层面的知识以及获取、理解和掌握知识的能力，加上感性层面的情感以及产生、感知和表达情感的能力。之所以民智成为一个候选者，原因是随着人类可利用算力的增加，高级 AI 的出现，生产自动化的普及，初级劳动力将失去价值，物质财富的生产可能进入“边际成本趋近于零”的状态，这意味着用来组织生产的资本将不再稀缺。到这个时候，真正重要的问题变成了“做什么”，以及“为什么要做”，民智就可能成为贡献最大的，处于组织核心的，可转化为其他生产要素的首位生产要素。这样一种未来非常值得期待，它意味着复辟官僚制度剥削手段的彻底失效。统治者可以控制人的身体，可以垄断土地的产出，可以勒索流动的资本，但是他们永远无法通过榨取来获得真正有创造力的智慧。民智具有强烈的去中心化属性，要想获得其最大化的产出，社会就必须提供自由，宽容和民主的环境。民智真正伟大之处在于：它是不可剥削的，当它受到侵犯的时候，即使自我毁灭，也不会被凶徒所得。如果日后真的将出现一段以民智为首位生产要素的时代，那正是消除各国苦难，促进人类更为紧密团结的大好时机。

另一个备选项就不会那么受欢迎了，那就是“能源”。当我们提及能源可能成为首位的生产因素时，人们或许很容易想到著名的卡尔达肖夫文明指数（Kardashev scale），实际上也确有关联。要想民智得到充分发挥，不管是生产的自动化，还是作为支持的庞大算力都需要惊人的能源供给。这意味着人类不可能永远指望着仅靠地球上所有的化石能源、核能，甚至加上地表太阳能也不够。我们总归需要走出地球这个摇篮，在星际尺度上完成对于能源的获取和利用。在某种程度上，能源将会取代曾经土地所具有的地位。过去土地作为具有强烈空间属性的生产要素（也是生产资料）之所以为人们所争夺，一个重要原因是土地是前工业时代古代人类获得能量的几乎唯一载体，土地产出粮食，粮食转化为人口。当人类科技水平达到一定程度，甚至能够充分利用太阳的能量的时候，土地的二维空间限制就被打破了。只要有了充足的能源，人类可以改造之前所不能提供经济产出的土地，甚至在太空建立现在无

法想象的定居点。能源某种意义上将会成为未来的三维“土地”，控制它就意味着控制了新世界的“造物权”。与民智不同，能源是可以被国家机器所垄断的，因为它具有极高的技术、资本门槛和物理集中性。普通大众的民智在能源垄断面前会丧失讨价还价的余地，因为即使是算力，甚至是生存环境的维持系统都会被能源拥有者所间接控制。这种未来在比较糟糕的情况下，会形成今天所谓“赛博朋克式”的世界。但是让人感到难以评说的是，即使是能源作为首位生产要素下被垄断的赛博朋克世界，也比明清帝国这种复辟官僚国家好了太多。因为后者的运作逻辑在于必须主动地折磨你，榨取你，必须用劳累、愚昧和恐惧把每一个人类个体变成干电池，驱使民力以维生。而在赛博朋克时代，AI和机器人使得榨取底层的民力变得没有意义了，统治者对被统治的态度将从“残酷的压榨”变成“冷漠的无视”。虽然听起来仍然很悲哀，但是确实在物理层面上终结了那个靠吃活人血肉来运转的历史周期律。

那么如何追求一个较好的未来，而避免那个较糟糕的呢？别忘了我们已经知道稀缺性，对于生产关系具有重要的调节作用。答案是呼之欲出的：要想确保民智在下一个时代充分显示其威能，我们就必须确保能源不处于一种极度短缺状态。只有能源技术取得爆炸式突破，能源完全不再稀缺时，人类的竞争焦点才会彻底转移到民智上，压迫和垄断才会失去物理学意义，我们这个物种有机会迎来自由的，全面的解放。如果能源技术迟迟不突破，而AI的发展又过快，那么能源就将处于一种稀缺状态，从而存在被垄断的风险。不过我想说的是，即使未来的风险已经能够被预见，也不意味着人类应该采取一种“远视主义”的政策，毕竟风险只是一种可能，未来的情况也无法在今天就被准确获知。为了一个“不确定的宏伟目标”就加强对全社会的管制很有可能在当下就带来灾难性的后果，因为这正是复辟官僚体制所喜欢的，就如同《流浪地球》所反映的中国式价值观一样。增强对大众的科普，政策上的扶持加上对科研的持续投入和赞助也能够加快新能源革命的那一天到来。总而言之，人们应该采用温和的办法，而不是强硬的行政手段。

当然，我们没有理由认为民智和能源代表了互斥的两条路线，实际上，他们很有可能各自统领了时间上先后的两个时代：首先民智成为首位生产要素，在这个时代，人类首先越过了卡尔达肖夫一型文明的门槛。在继续向二型文明的演化过程中，能源将成为首要的生产要素。这也是一个令人欣喜的结果，因为民智时代将给人类带来又一轮的解放，这使得在能源主导的星际航行时期我们不必再背着迄今为止的历史包袱前行，人们将会比今天更加理性、更有智慧地看待政治。这样甚至会出现一个有趣的情形，如果我们假定接下来真的要经历民智和能源两个时代，那么，从古至今到未来就出现了“初级劳动力、土地、资本、民智（某种意义上是高级劳动力）和能源”五个生产要素分别主导的时代。如果这种预测成真的话，那么我们的确处于“二阶段”的开始，一个毋庸置疑的转捩点上。不难看出，初级劳动力和民智都是和人自身有关的创造力，随后的土地和能源都是自然界的物理属性，而在这两个阶段中，由资本进行了“润滑”。我们甚至还可以大胆预测，将来很可能会有一个宇宙级别的

“大扩张”时代。当人类用亚光速的飞行器向宇宙深空推进的时候，必然会形成一种晚期智人扩散时的情形，那就是就物种整体上的扩散是迅速的，但是个体（定居点）在交流上是迟缓的。尽管人类可以通过代际之间的接力扩张，不断的利用较早建立的据点，把自身的存在迅速（例如1万年到100万年，相对于地质时间尺度仍然是比较快的）遍布于一个广大的宇宙空间，但是各个宇宙定居点却难以进行双向交流，他们很可能会相对独立发展。因为有光速这个硬性限制，将来会不会有宇宙尺度的早期国家呢？出于降低管理成本的必要性，会不会有宇宙尺度的“能源封建化”呢？如果在更加遥远的未来出现一些现在无法料及的超技术（比如虫洞），那么宇宙尺度的能源封建化人类文明还会进一步演化，这是多么瑰丽的想象啊！

最后，我们还有一点可以比较确定地预料，那就是传统意义上的阶级正在消失。这并不是说阶级本身消失了，而是说保持这个定义的意义在消失。我们在描述人类过往的历史中，之所以强调人和社会的阶级属性，是因为不同阶级之间所处的相对地位不同，在社会生产中承担的功能不同，因此会有特定的互动模式。特别地，过往的阶级社会总有一个主要的剥削阶级和被剥削阶级。这是因为哪怕一个群体控制了首位生产要素进而主导了社会生产，他总归要利用民力进行劳动生产，经济上的不平等最终带来了社会上的不平等。但是随着人工智能和自动化的全面铺开，大部分的劳动生产将不再由人完成，而人类所继续负责的创造性思考和情感活动，又不是通过剥削能够获得的。这毫无疑问将宣告过去那种比较分明的阶级对立的社会将迎来终结。在未来，人们很可能已经无法准确的指出在经济上进行统治的阶级和被统治的阶级了。随着阶级在人类社会中扮演角色的弱化，之前社会形态过渡时期支撑帝国崛起的阶级更替也不会再出现。凡此种种变化将给未来人类的发展造成哪些影响，现在仍不可知，但是对我来说它们至少说明了一点，那就是我的文稿在今天这个时代被写出来的确意味着一段漫长岁月的终结，这或许也算是上天的旨意吧。